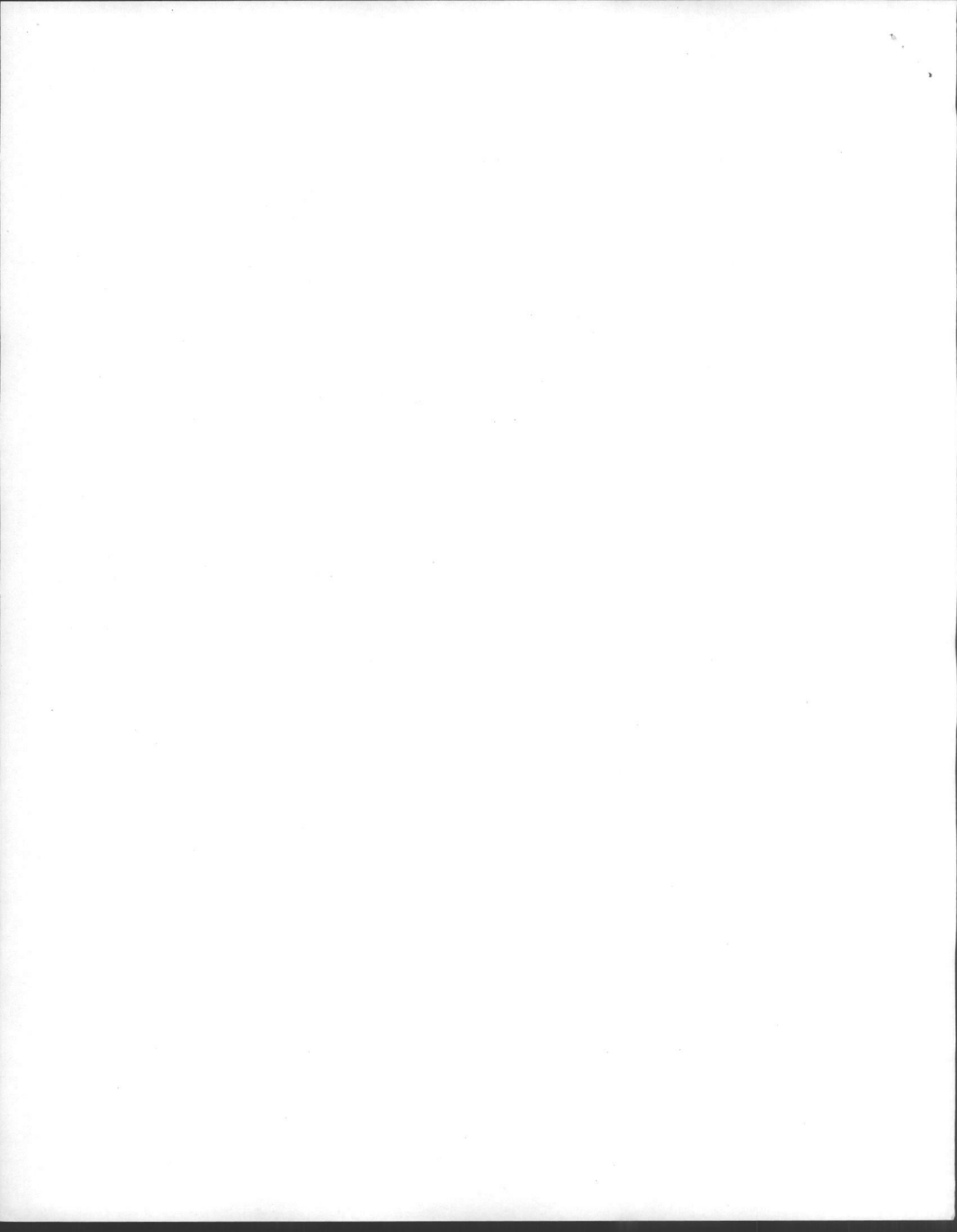


ALGEBRA I PRACTICE TEST



Part II

Answer all 6 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit.

25 Solve the equation below algebraically.

below algebraically.

$$6 - \frac{2}{3}(x + 5) = 4x$$
$$6 - \frac{2}{3}x - \frac{10}{3} = 4x$$
$$2.67 - \frac{2}{3}x = 4x$$
$$+ \frac{2}{3}x \quad + \frac{2}{3}x$$

$$\frac{2.67}{4.67} = \frac{4.67x}{4.67}$$
$$x = 0.57143$$

26 $F = 4x^2 - 3x + 1$ and $G = -2x^2 + 5$. Find the trinomial that represents H , if $H = 3G - F$.

$$H = 3G - F \rightarrow H = 3(-2x^2 + 5) - (4x^2 - 3x + 1)$$

$$H = -6x^2 + 15 - 4x^2 + 3x - 1 \rightarrow H = -10x^2 + 3x + 14$$

27 Algebraically determine the zeros of the function $f(x) = x^2 - 4x + 3$.

x^2	$-3x$
$-x$	3

$$= -4x$$

~~$$(x-3)(x-1) = 0$$~~

$$(x-3)(x-1) = 0$$

$$x-3=0 \rightarrow 3-3=0 \rightarrow x=3$$

$$x-1=0 \rightarrow 1-1=0 \rightarrow x=1$$

$$x=3, x=1$$

- 28 In February 2021, the state of Texas saw one of the worst snowstorms in its history. Snowfall totals recorded in Dallas, TX are shown in the table below.

Dallas, TX	
Time	Snow (inches)
1 a.m.	1
3 a.m.	5
6 a.m.	11
12 noon	33
3 p.m.	36

Handwritten calculations and annotations:

- From 1 a.m. to 6 a.m. (5 hours): $\frac{11 \text{ inches} - 1 \text{ inch}}{5 \text{ hours}} = 4.55 \text{ inches per hour}$
- From 6 a.m. to 12 noon (6 hours): $\frac{33 \text{ inches} - 11 \text{ inches}}{6 \text{ hours}} = 8.89 \text{ inches per hour}$
- From 12 noon to 3 p.m. (3 hours): $\frac{36 \text{ inches} - 33 \text{ inches}}{3 \text{ hours}} = 3 \text{ inches per hour}$

Which of the following intervals had the greatest rate of snowfall, in inches per hour? 1 a.m. to 12 noon or 6 a.m. to 3 p.m. Justify your answer.

The interval 6 a.m. to 3 p.m. had the greatest rate of snowfall. 6 a.m. to 3 p.m. was a 9 hour interval with 80 inches of snow, which is 8.89 inches of snow per hour. 1 a.m. to 12 noon was an 11 hour interval with 50 inches of snow, which is 4.55 inches of snow per hour. 8.89 inches of snow per hour is a greater rate of snowfall compared to 4.55 inches of snow per hour.

- 29 The gravitational force between two objects can be calculated using the formula $F_g = \frac{GM_1M_2}{r^2}$. In the formula, G is called the gravitational constant. The variables M_1 and M_2 represent the masses of the two objects between which the force is being measured. Lastly, r is the distance between the objects. Solve for the positive value of r , in terms of the other variables.

$$\begin{aligned} F_g &= \frac{GM_1M_2}{r^2} \rightarrow \cancel{r^2} \cancel{F_g} \\ \downarrow r^2 &= \frac{GM_1M_2}{F_g} \rightarrow r = \sqrt{\frac{GM_1M_2}{F_g}} \\ r &= \sqrt{\frac{GM_1M_2}{F_g}} \rightarrow r = \sqrt{r^2} \\ r &= \sqrt{r^2} \xrightarrow{\text{cancel all}} \boxed{r=r} \end{aligned}$$

- 30 Four Seasons Landscaping is creating a rectangular flower bed at a customer's home. The width of the flower bed will be half of the length. The total area of the flower bed is 38 square feet. Write and solve an equation to determine the width of the flower bed to the *nearest tenth* of a foot.

$W = \text{width}$
 $L = \text{length}$

$$W = \frac{L}{2}$$

$$A = L \cdot W \rightarrow 38 \text{ square feet} = L \cdot \frac{L}{2}$$

$$38 = L \cdot \frac{L}{2} \rightarrow \frac{38}{L} = \frac{L}{2}$$

$$\frac{38}{8.72} = \frac{8.72}{2} \rightarrow 4.36 = 4.36$$

$$\boxed{\cancel{W = 4.35}} \rightarrow \boxed{W = 4.36}$$

Part III

Answer all 4 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.

- 31 A couple planning a backyard wedding is researching different tent rental companies for their event. They contact two different companies. Buffalo Party Rental charges a \$50 set-up fee, and \$5 per square foot of tent space. Tent Genie charges a \$25 set-up fee, and \$6 per square foot of tent space.

Write an equation for Buffalo Party Rental that could be used to determine the total cost B , when x square feet of tent space is ordered. Write a second equation for Tent Genie that could be used to determine the total cost T , when x square feet of tent space is ordered.

Buffalo Party Rental

$$y = mx + b$$

~~$$y = 5x + 150$$~~

$$B = 5x + 50$$

$m = \$5$ per square foot

$b = \$50$ set up fee

Tent Genie

$$y = mx + b$$

$$T = 6x + 25$$

$m = \$6$ per square foot

$b = \$25$ set up fee

Determine algebraically and state the *minimum* square feet, rounded to the *nearest square foot*, of tent space that must be ordered for Buffalo Party Rental to be the cheaper option.

$$B < T \rightarrow 5x + 50 < 6x + 25$$

$$\begin{array}{r} 5x + 50 < 6x + 25 \\ -25 \quad -25 \\ \hline 5x + 25 < 6x \end{array}$$

$$\begin{array}{r} 5x + 25 < 6x \\ -5x \quad -5x \\ \hline 25 < x \end{array}$$

$$25 < x \rightarrow \text{minimum} = 26 \text{ square feet}$$

$$5(25) + 50 = 175 = 6(25) + 25$$

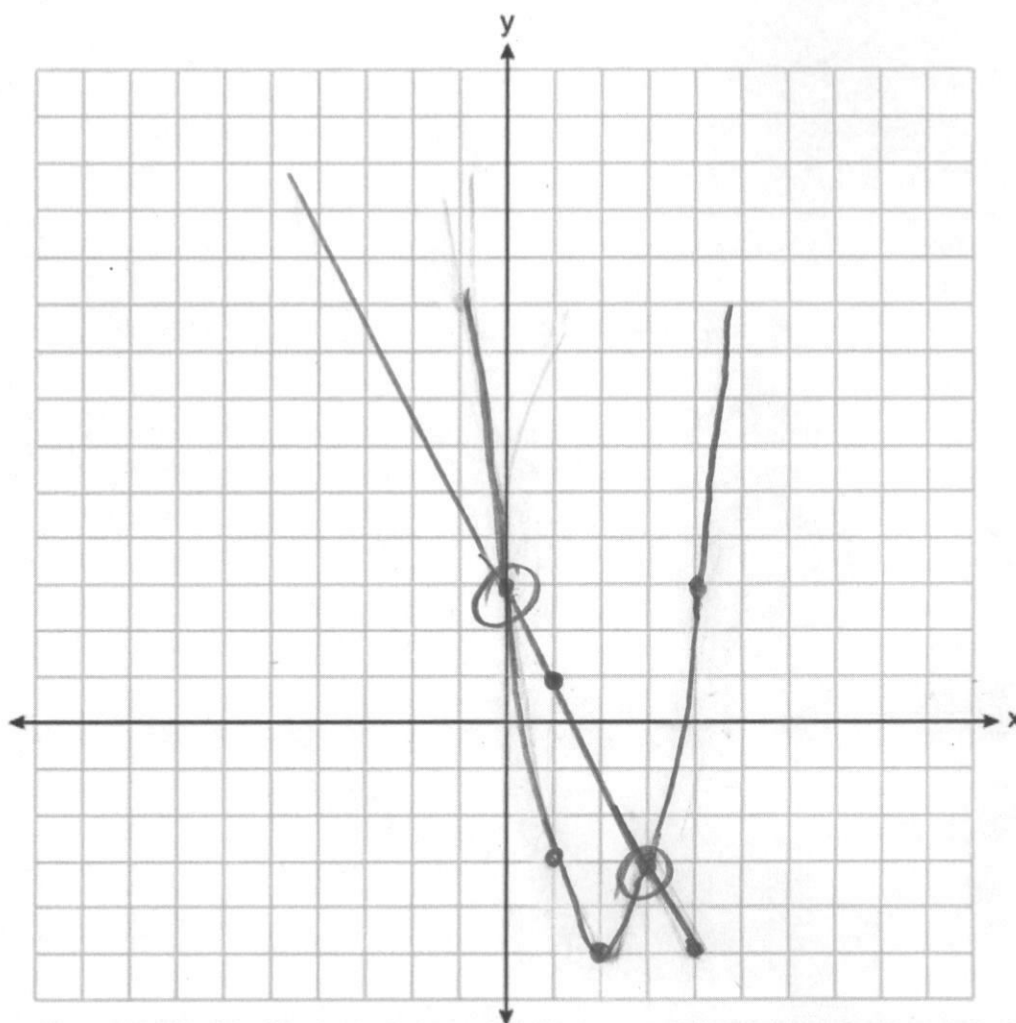
$$5(26) + 50 = 180$$

$$6(26) + 25 = 181$$

32 Graph the two functions on the coordinate plane below.

$$f(x) = 2x^2 - 8x + 3$$

$$g(x) = -2x + 3$$



Find all values of x for which $f(x) = g(x)$.

$(0, 3)$ $(3, -3)$

- 33 The table below shows the number of hours ten different scholars at Marion P. Thomas Middle School spent studying and their scores on a recent history test.

Hours Spent Studying (x)	0	1	2	4	4	4	6	6	7	8
Test Scores (y)	35	40	46	65	67	70	82	88	82	95

Write the linear regression for this data set. Round all values to the nearest hundredth.

$$y = ax + b$$

$$a = 7.98 \quad b = 33.80$$

$$y = 7.98x + 33.80$$

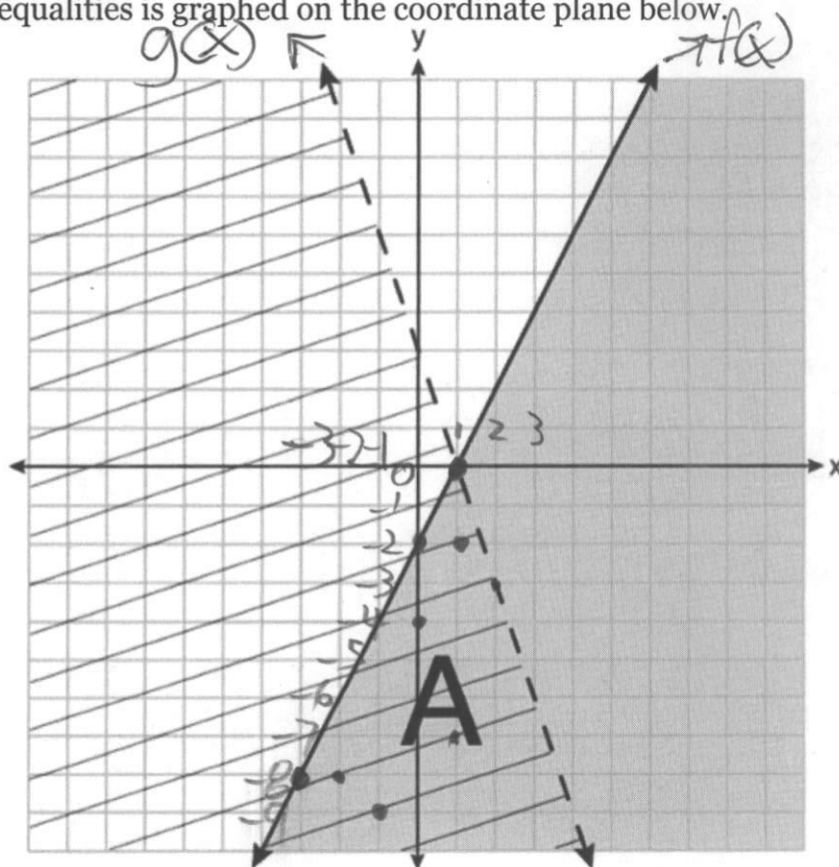
State the correlation coefficient rounded to the nearest hundredth.

$$r = 0.98$$

Explain what the correlation coefficient suggests in the context of the problem.

The correlation coefficient suggests that this data set has a strong correlation (r is closer to 1).

34 A system of inequalities is graphed on the coordinate plane below.



Write the system of inequalities that is represented by the graph above.

$(1, 0)$ $(0, -2)$ $(1, -2)$
 $(2, -3)$ $(0, -4)$ $(1, -7)$
 $(3, -8)$ $(2, -8)$ ~~$(1, -9)$~~ $\rightarrow (-1, -9)$

What does region A represent?

~~The~~ Region A represents the solution. In Region A, both inequalities meet. Since both inequalities meet at Region A, it represents the solution.

What does the gray region represent?

The gray region represents the solutions to line $f(x)$.

Part IV

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.

- 35 The CoinStar machine at the local supermarket is broken. Instead of providing the exact number of each type of coin that is deposited, the machine can only display the total number of coins and total value. The machine shows 90 coins, with a value of \$17.55.

If Mr. Rogers only placed dimes and quarters in the machine, write a system of equations in two variables, or an equation in one variable that could be used to model this situation.

~~Q = 25c~~

$\$0.25 = 1 \text{ Quarter}$
 $\$0.1 = 1 \text{ Dime}$

~~$\$0.25Q + \$0.1D = \$17.55$~~

$\$0.25Q + \$0.1D = \$17.55$

~~$C = 90$~~ $Q = \# \text{ of quarters}$
 $D = \# \text{ of Dimes}$

Using your equation or system of equations, algebraically determine the number of quarters Mr. Rogers deposited in the machine.

$D = 33$

$\$17.55 = \$0.25Q + \$0.1D$

$\$0.25Q = \$17.55 - \$0.1D$

$\$0.25Q = \$17.55 - \$0.1(33)$

$\$0.25Q = \14.25

$\frac{\$0.25}{\$0.25} \quad \frac{\$14.25}{\$0.25} \quad Q = 57 \text{ quarters}$

Mr. Rogers' neighbor replaced each one of his dimes with a quarter. If he uses all of his coins, determine if Mr. Rogers would have enough money to buy a cardigan priced at \$20.98 if he must also pay an 8% sales tax. Justify your answer.

$$33 \text{ Dimes and } 57 \text{ Quarters} \rightarrow 90 \text{ Quarters}$$
$$90 \text{ Quarters} = \$0.25 \cdot 90 = \$22.5$$

$$\text{Cost of Cardigan} = \$20.98 + 8\% \text{ tax}$$

$$\text{Cost of Cardigan} = \$20.98 \cdot 1.08 = \$22.66$$
$$\$22.66 > \$22.5$$

Mr. Rogers would not have enough money to buy a cardigan. Since Mr. Rogers' neighbor replaced Mr. Rogers' dimes with quarters, he now has 90 quarters which equals \$22.5. The total cost of the cardigan with tax would be \$22.66. The total cost of the cardigan is greater than the total amount of money Mr. Rogers has. This means Mr. Rogers ~~will~~ will not have enough money to afford the cardigan.